Global CO_2 Emissions: Trends, Regional Shifts, and Per Capita Inequalities

A data-driven overview from the Industrial Revolution to 2024

Sources: Global Carbon Budget 2025 · NASA / NOAA Mauna Loa Observatory

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429

parts per million

Latest measurement
NOAA Mauna Loa Observatory, February 2026

Pre-industrial baseline
~280 ppm (more than 50% lower)

Rate of change
A change that took thousands of years at the end of the last ice age has occurred in roughly two centuries, linked primarily to burning coal, oil, and natural gas.

Source: climate.nasa.gov/vital-signs/carbon-dioxide/

Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

1950

6 Bt

Billion tonnes of CO_2

1990

20+ Bt

Nearly quadrupled

Today

35+ Bt

Annual fossil fuel + industry

Emissions growth has slowed in recent years, but has not yet peaked. The trajectory from 1750 to 2024 shows accelerating release of carbon into the atmosphere, driven by industrialization and energy demand.

Reference: figure_2.jpg (Global emissions trajectory line chart)

Source: Global Carbon Budget 2025 / Our World in Data

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Country / Region	Annual CO_2 (billion t)	Share of Global	Trend
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
United Kingdom	~0.3	~0.9%	Declining
Brazil	~0.5	~1.4%	Roughly stable
France	~0.3	~0.8%	Declining

Reference: figure_3.jpg (Annual CO_2 emissions by country)

China's steep rise post-2000 contrasts with the US peak near 6 Bt around 2005 and subsequent decline. India is now the third-largest emitter and rising fast.

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900

>90%

Europe + United States

The industrialized West dominated global emissions at the dawn of the 20th century, with Europe and the US accounting for the vast majority of CO_2 output.

2024

~50%

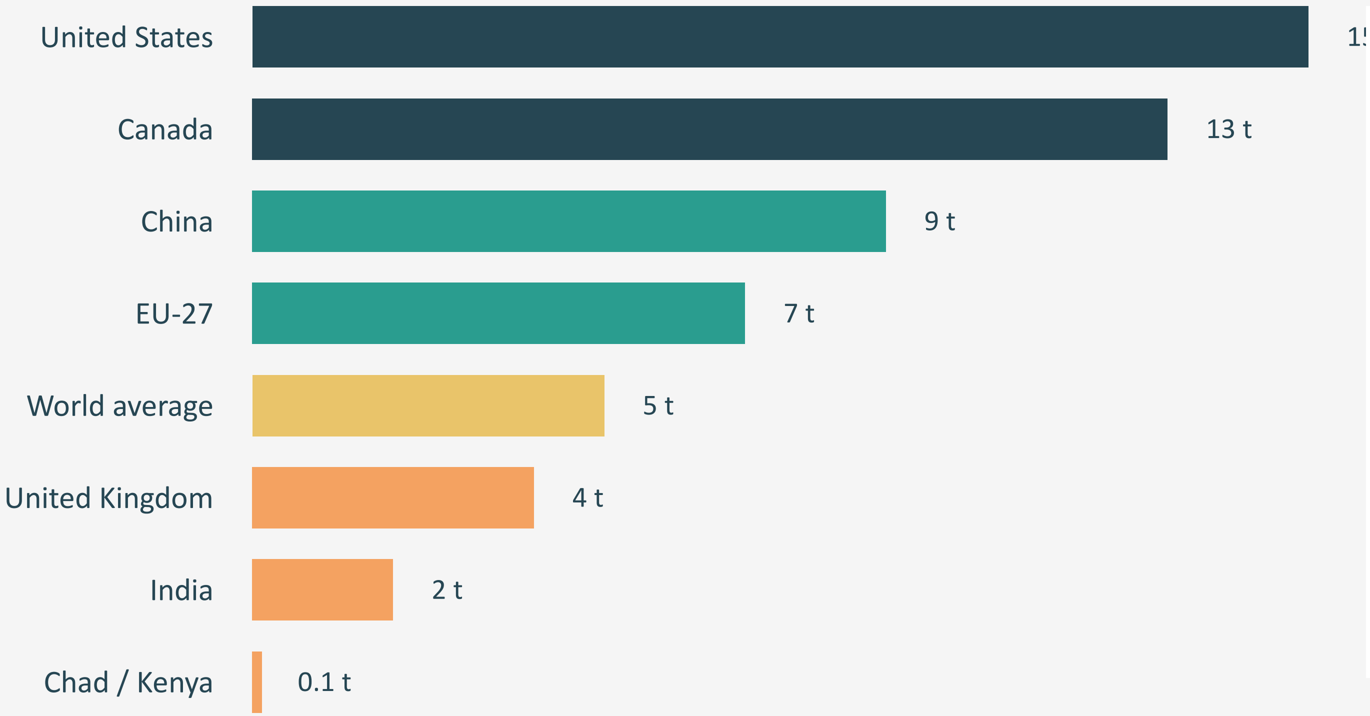
Asia's share

Asia — home to nearly 60% of the world's population — now produces roughly half of all emissions. The US and Europe combined account for less than one-third. Africa and South America each contribute only 3–4%.

Visual: Stacked area or side-by-side comparison showing dramatic regional shift from 1900 to 2024

Source: Global Carbon Budget 2025 / Our World in Data

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



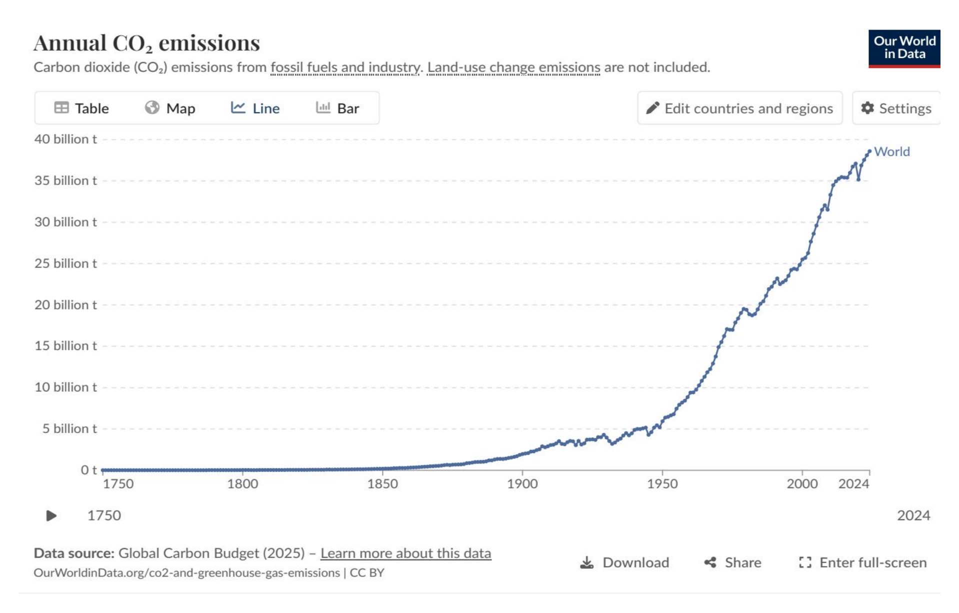
150×

gap between highest and lowest per capita emitters

An American emits in under two days what a person in Chad or Niger emits in a full year.

Reference: figure_1.jpg (Per capita CO_2 emissions over time)

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase



Reference: figure_4.jpg (Annual percentage change in CO_2 emissions, 2024)

Regional Patterns

- Declining (−1% to −5%)
  United States and much of Europe
- Increasing (+2% to +10%)
  Russia, parts of Southeast Asia, several African nations
- Very large increases (>10%)
  One or two countries in South America

The 2024 map reveals a mixed global picture: mature economies are decoupling growth from emissions, while emerging economies continue to rise.

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

	France	United Kingdom	Germany	Netherlands
Nuclear share	High	Moderate	Low	Very low
Renewables share	Growing	Growing	Growing	Growing
Fossil fuel share	Lower	Lower	Higher	Higher
Per capita CO_2	~4 t	~4 t	Higher	Higher

Countries with similar living standards can differ significantly in per capita emissions. France and the UK emit roughly half the per capita CO_2 of Germany and the Netherlands, largely because of a higher nuclear and renewable share in their energy mix.

Source: Global Carbon Budget 2025 / Our World in Data

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

1. Global CO_2 emissions now exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.
2. China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt).
3. Per capita inequalities remain extreme — a 150× gap separates the highest emitters (US, Australia at ~15 t/person) from the lowest (Chad, Niger at ~0.1 t/person).
4. The US and Europe are declining in annual emissions, but historical cumulative responsibility remains large.
5. Policy and energy mix choices — not just prosperity — determine per capita outcomes.

Sources: Global Carbon Budget 2025 · NASA / NOAA · Our World in Data